

# Cave detection using the Self-Potential-Surface (SPS) technique on a karstic terrain in the Jura mountains (Switzerland)

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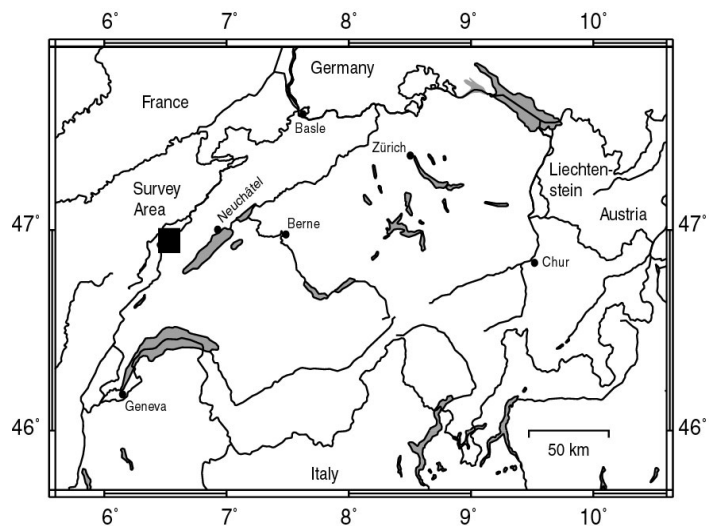
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## 1 Introduction

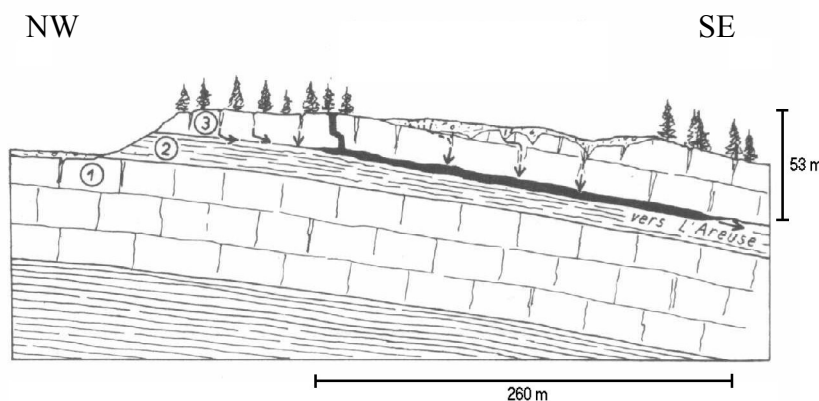
Originally, Aubert & Atangana (1996) developed the Self-Potential-Surface (SPS) technique for hydrogeological exploration of volcanic areas. Provided that two conditions are satisfied, they experimentally found a linear relation between the range of negative Self-Potential (SP) anomalies and the thickness of the unsaturated zone. The first condition is a strong contrast between the resistivity of the unsaturated zone and the resistivity of the substratum and the water-saturated zone. The second condition is the homogeneity of the unsaturated zone. The qualitative interpretation of SP maps indicates that the lines of maximum negative values correspond to the drainage axes and the lines of minimum negative values to the boundaries between two watersheds (Jackson & Kauahikaua (1987)).

We expected similar conditions to be valid in carbonate aquifers. Especially in a karstic cave, the resistivity contrast created by the air-layer must be significant, so that the SPS technique can be used to detect these structures.

Motivated by this hypothesis, an experiment was conducted in co-work with the Centre of Hydrogeology Neuchâtel (CHYN).



**Figure 1:** Map of Switzerland with survey area location



**Figure 2:** Schematic geological cut of the Chez le Brandt Cave area; 1&3=karstified Sequanian/ Argovian Limestone; 2=Marl (after Müller, 1981)

Argovian (Fig.2) with a length of about 260m. A marl layer of this sequence allowed the development of a small subterranean river. Tectonic features like faults are visible inside the cave (Müller, 1981). These strike directions coincide with the local stress parameters that controlled fracturing and karstification in the Swiss Folded Jura (main shearing =  $N0^\circ$ ,  $\sigma_1 = N130^\circ$ - $N150^\circ$ ,  $\sigma_2 = N40^\circ$ - $N50^\circ$ ).

## 2 Survey area

The field experiment has been carried out over the cave of Vers-Chez-Le-Brandt (Bosch & Gurk, 2000) near to France in the folded limestone of the Swiss Jura Mountains, canton Neuchâtel (Fig. 1). In this area Mesozoic limestone and marl crop out, covered by thin Quaternary deposits. The cave is developed in direction of about  $N140^\circ$  in karstified limestone of the upper Sequanian/

### 3 SPS method

Figure 3a represents the SPS model for a subhorizontal layered structure in karst. According to Aubert et al. (1990) and Aubert & Atangana (1996), three specific conditions shall be satisfied in this model:

- 1) A high ratio between the resistivity of the unsaturated zone (A) and the resistivity of the impermeable zone (B)
- 2) Zone (A) is composed out of porous and homogenous medium
- 3) The resistivity distribution of zone (A) is homogenous

The interface (C) is the contact between zone (A) and zone (B). Under the above mentioned conditions, Aubert et al. (1990) and Aubert & Atangana (1996) give the following experimental linear relation:

$$d_{(x,y)} = \frac{V_{(x,y)}}{K} + d^{\circ} \quad (1)$$

$d_{(x,y)}$  is the thickness of the unsaturated zone (A) and  $V_{(x,y)}$  is the range of SP anomaly measured at the surface.

$K$  is a coefficient in Volt/Meter,  $E^{\circ}$  is the known thickness of zone (A) below the SP reference point  $R$ , where  $V_{(R)} = 0$  Volt by convention.

The interface (C) would then be an equipotential surface, called the Self-Potential-Surface (SPS).

$K$  is a function of various parameters including the average resistivity of zone (A) and the resistivity of zone (B). When the SPS is sub-horizontal,  $K$  defines the topographic effect in SP and we can derive  $K$  by:

$$K = \frac{dV}{dh}, \quad (2)$$

i.e. by the vertical SP gradient from the reference point  $R$  down to the surface of zone (B):

$$K = \frac{dV_{(R)}}{d^{\circ}}, \quad (3)$$

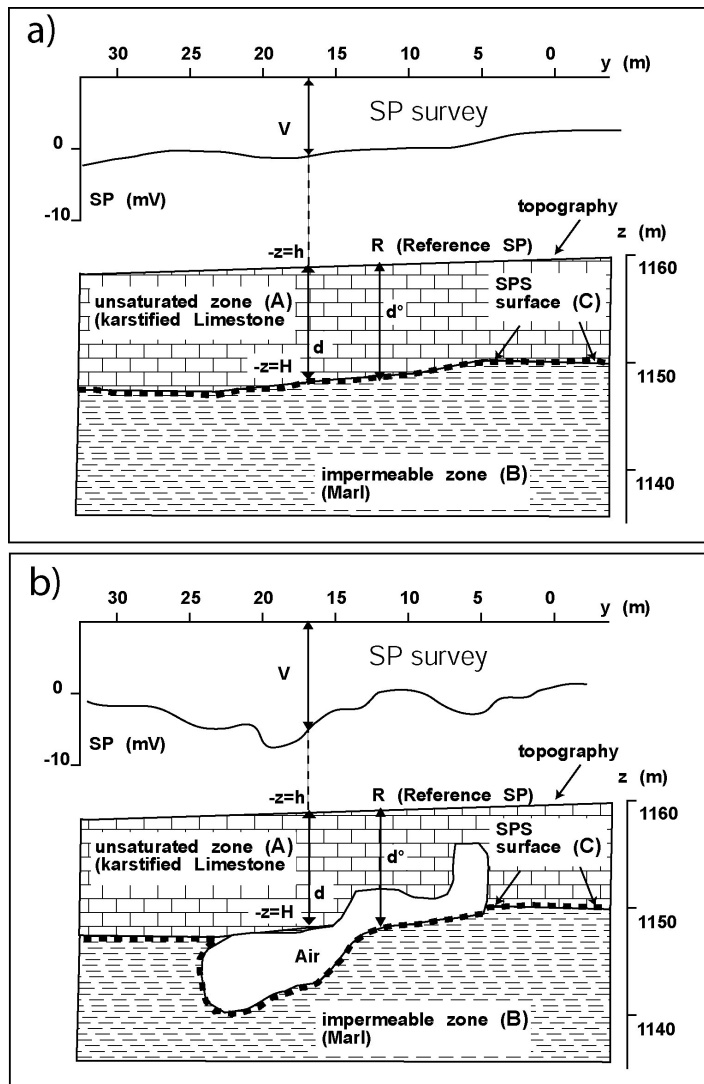
$K$  can be determined from borehole information or laboratory measurements of rock samples of zone (A).

If the field topography  $h_{(x,y)}$  and the SP mapping  $V_{(x,y)}$  are known, we can then calculate the topography of the SPS:

$$H_{(x,y)} = h_{(x,y)} - d^{\circ} - \frac{V_{(x,y)}}{K} \quad (4)$$

The SPS topography will give the topography of the impermeable zone (B).

A cave will significantly change the potential distribution in the structure and therefore the SP mapping at surface. In our working model for the field experiment (Fig. 3b), a possible equipotential surface is thought to align with the bottom of the cave. Another possible equipotential surface can be considered to follow the roof of the cave.



**Figure 3a:** SPS model for a sub-horizontal layered structure in karst

**Figure 3b:** SPS model for a sub-horizontal layered structure in karst with a cave

#### 4 Field experiment

The experiment was carried out over a shallow part (depth between 5 to 12 m) of the cave (Fig. 5).

##### *Positioning of the reference Electrode*

Besides the careful sampling of the SP anomalies, the accurate estimate of  $K$  is crucial for the method. One of the main problems in a cave is to determine  $E^\circ$  and the plumb line between the SP reference point  $R$  to the vertical electrode inside the cave.  $d^\circ$  must be taken from maps and crosscuts.

To delineate the projection of the position of the vertical electrode (placed inside the cave) to the surface we now use a triangulation method:

At the chosen position of the vertical electrode in the cave we set up a portable VLF transmitter of 13.5 kHz (Fig. 6) and triangulating its projection to the surface with the RMT (Radiofrequency MagnetoTelluric) equipment (Fig. 7) developed at CHYN. This method will give the position  $R$  for the reference electrode of the SP mapping at the surface (Fig. 3a and 3b).



**Figure 4:** Self-Potential equipment "Type Münster"

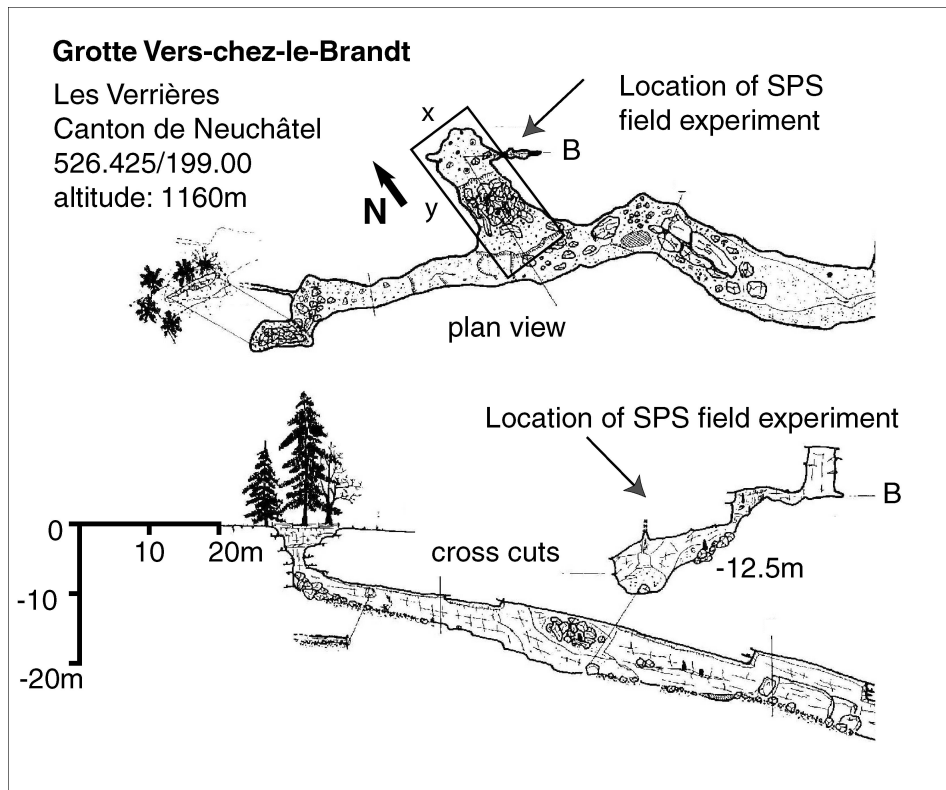
##### *Vertical SP gradient ( $K$ )*

Estimates of  $K$  were obtained by placing an electrode in the bottom of the cave (zone (B)) beneath reference point  $R$  and sampling the vertical SP gradient between this electrode and the reference electrode at the surface.

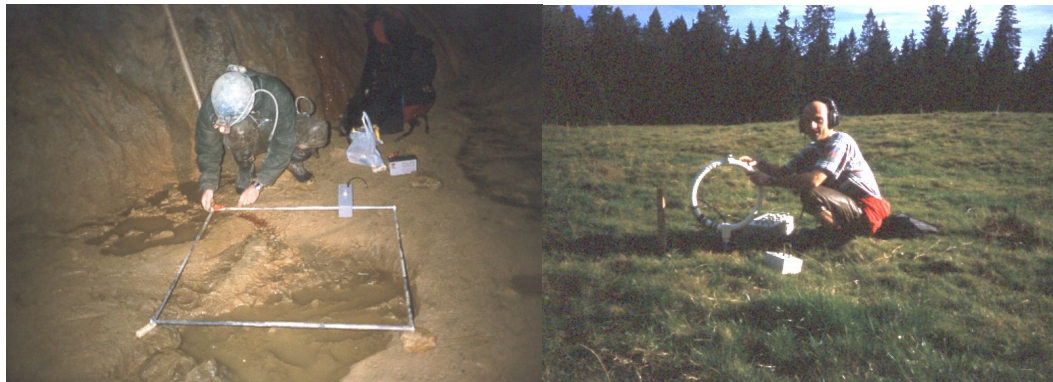
$K$  was found in this experiment to be  $-10.3 \text{ mV}/15\text{m}$ .

##### *Topography and SP mapping at the surface*

We sampled topographic and SP data at the surface in a  $10\text{m} \times 20\text{m}$  area ( $x \rightarrow$  East,  $y \rightarrow$  South). SP data were collected every meter using saturated  $\text{Cu-CuSO}_4$  electrodes arranged on a spade stick (Fig. 4). All SP values are sampled in addition to the surface reference electrode at position  $R$ .



**Figure 5:** The experiment site and part of the cave Vers-Chez-Le-Brandt



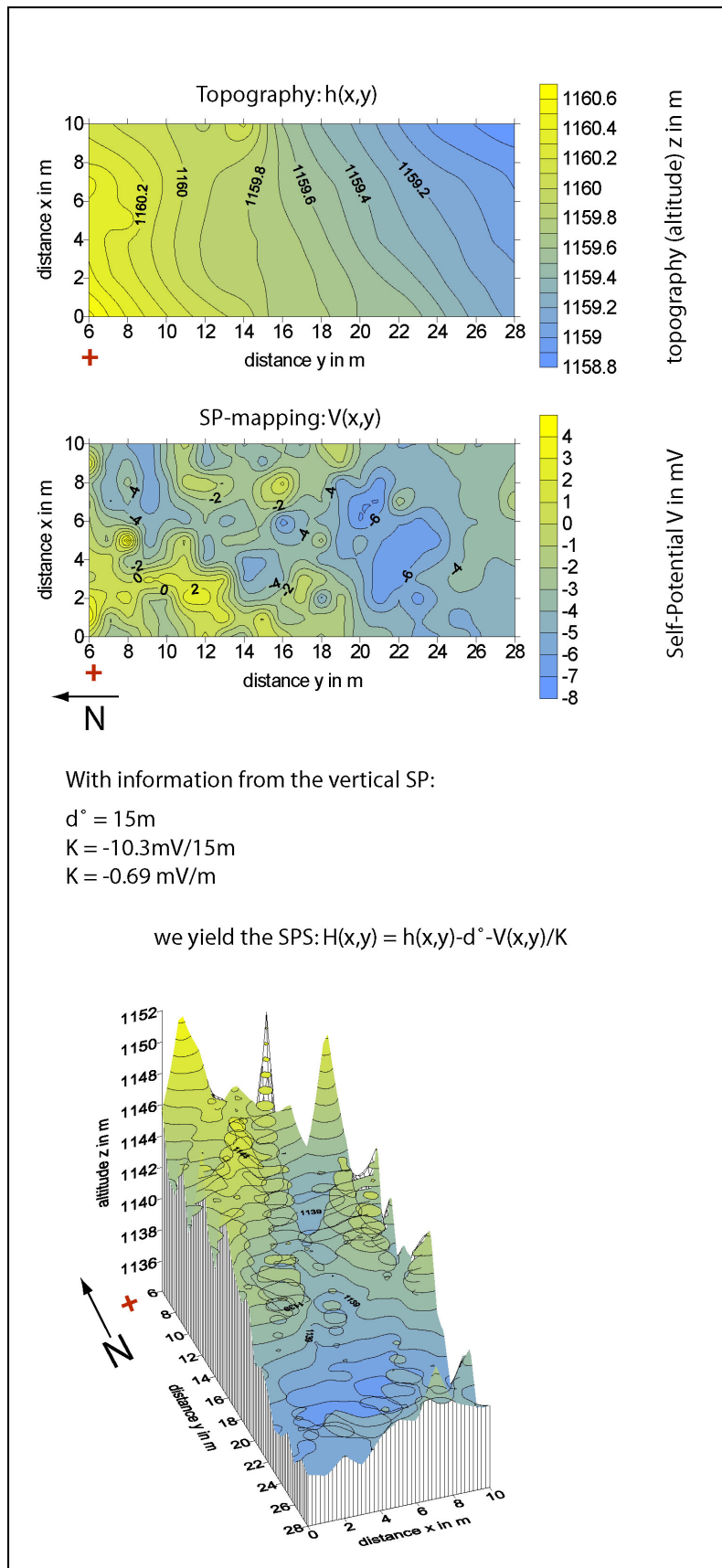
**Figure 6 (left):** Portable VLF transmitter (grey box) of 13.5 kHz on the bottom of the cave, the vertical electrode will be placed in the centre of the rectangular coil

**Figure 7 (right):** CHYN's RMT receiver and horizontal coil above the cave to triangulate the projection of the transmitter position to the surface. This will give the SP reference point  $R$ .

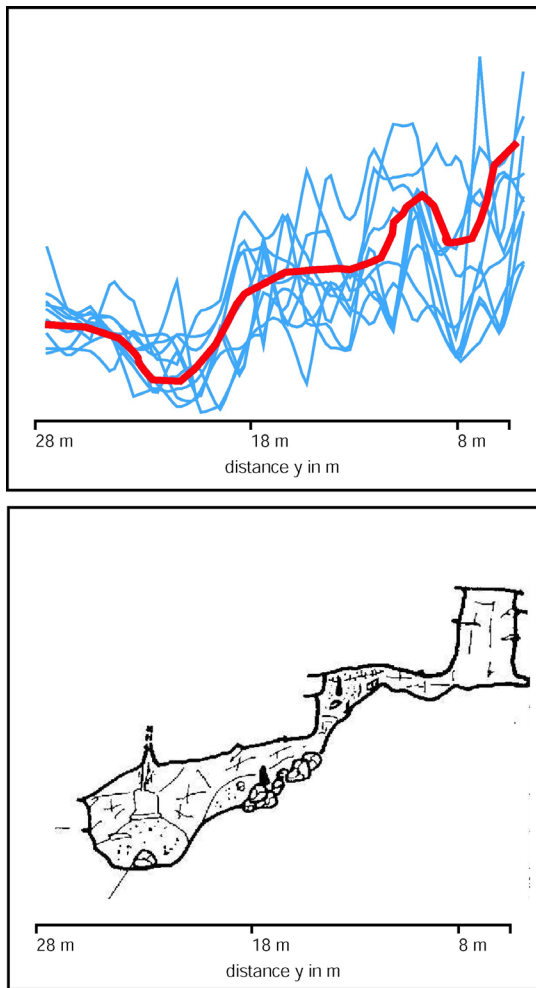
## 5 Results

The findings of the experiment are presented in Figure 8. First of all we find that the SP-mapping is decoupled from the surface topography, indicating that the SP is created by internal features of the subsoil. The SP data varies between values of  $-8$  to  $+4$  mV, which is found to be a typical range for SP data on karst areas in the Jura mountain (Bosch & Gurk, 2000). The SP mapping itself allows drawing conclusion about the position of the cave: Maximum negative SP values at  $y = 20-28$ m are indicating the main cave and main drainage axis to be East – Southeast with a branch (minimum negative values) that leads towards the North.

However, the valence of the SP data with the topographic information and the vertical SP gradient ( $K$ ) gives a detailed view insight the cave's topography. The estimate of  $d^\circ$  and  $K$  was sufficient to give a comparable topography of the bottom of the cave (Fig. 9).



**Figure 8:** Topography, SP-mapping and the resulting SPS over the cave Vers-Chez-Le-Brandt. For spatial position of the survey see Fig. 5 (x-> East, y -> South).



**Figure 9:** Comparison of SPS slices of Fig 8 ( $x=1, 2, \dots, 10$ , blue, respectively thin lines) and its visual average (red, respectively thick line) with the crosscut based on speological data of the cave taken from Fig. 5.

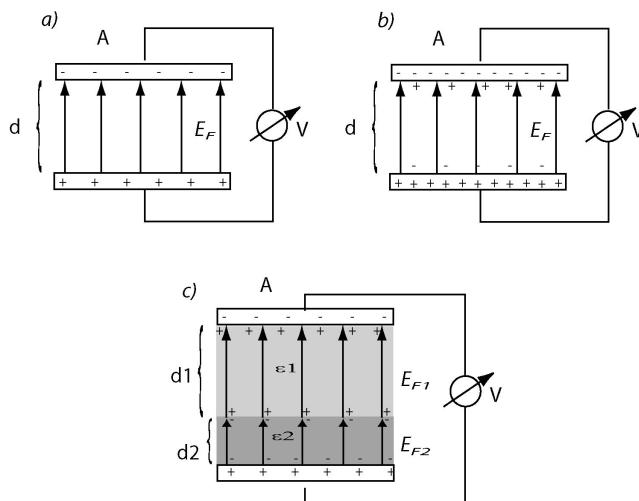
## 6 Discussion

By definition, the SPS technique does not account for any inhomogeneities, such as a cave. Theoretically, the method should have failed. However, to explain our experimental observation at surface we divide the problem into an active and static part. The active part comprises the mechanism that creates a charge accumulation along the SPS, whereas the passive part explains how this charge distribution is mirrored to the surface. The latter question can be qualitatively explained in terms of a capacitor with various dielectric media:

*Static model for a subhorizontal layered structure without a cave*

We state that the  $K$  factor, as defined in the original idea by Aubert et al. (1990) and Aubert & Atangana (1996) for a subhorizontal layered structure of homogenous media, is similar to the electric field strength in a capacitor (Fig. 10 a)):

$$E_F \cong K. \quad (5)$$



$E_F$  electrical field strength  
 $A$  area  
 $Q$  charge  
 $D$  displacement of charge  
 $d$  distance  
 $\epsilon_0$  electrical permittivity of vacuum  
 $(\frac{1}{36\pi} 10^{-9} \frac{As}{Vm})$   
 $\epsilon_r$  relative electrical permittivity  
 $(\epsilon_{\text{limestone}} = 8 - 12, \epsilon_{\text{water}} = 80, \epsilon_{\text{air}} = 1)$   
 $V$  potential  
 (Values taken from Landolt & Börnstein (1952))

**Figure 10:** a) Capacitor without and b) capacitor with a dielectric media. c) Capacitor with a series of dielectric media.

For a given charge distribution

$$\frac{Q}{A} = D = \epsilon_0 \epsilon_r E_F, \quad (6)$$

the electric field strength is defined as:

$$E_F = \frac{V_{(x,y)}}{d_{(x,y)}} \quad (7)$$

From this equation we obtain the potential:

$$V_{(x,y)} = E_F \cdot d_{(x,y)}, \text{ respectively } V_{(x,y)} = K \cdot d_{(x,y)}. \quad (8)$$

Similar to a capacitor, the charge on its boundaries is depending on the polarisation effect in a given dielectric medium between these boundaries. In a carbonate aquifer, the electrical permittivity of limestone enhances the charge by a factor 8 – 12:

$$\frac{Q}{A} = D = \epsilon_0 \epsilon_{\text{limestone}} K \quad (9)$$

With  $K = \frac{D}{\epsilon_0 \epsilon_{\text{limestone}}}$  we get a measure for the thickness of the structure:

$$d_{(x,y)} = \frac{V_{(x,y)}}{D} \epsilon_0 \epsilon_{\text{limestone}}. \quad (10)$$

And the potential is given by:

$$V_{(x,y)} = \frac{d_{(x,y)}}{\epsilon_0 \epsilon_{\text{limestone}}} D. \quad (11)$$

*Static model for a subhorizontal layered structure with a cave and water*

This model is characterised by a series of capacitors with different dielectric media:

$$D_1 = \epsilon_0 \epsilon_1 E_{F1}, D_2 = \epsilon_0 \epsilon_2 E_{F2}, \dots, D_i = \epsilon_0 \epsilon_{i2} E_{Fi} \quad (12)$$

Since the charge distribution at the boundaries are equal:

$$D_1 = D_2 = \dots D_i = D. \quad (13)$$

The resulting potential of a multiple layered structure is:

$$V'_{(x,y)} = \frac{D}{\epsilon_0} \left( \frac{d_1}{\epsilon_1} + \frac{d_2}{\epsilon_2} + \dots + \frac{d_i}{\epsilon_i} \right) = \frac{D}{\epsilon_0} \sum \frac{d_i}{\epsilon_i}. \quad (14)$$

To estimate the effect of these structures on our field experiment, we calculate the resulting potential for three specific models (Figure 11):

Model A: *limestone*

$$V'_{(x,y)} = \frac{D}{\epsilon_0} \left( \frac{d_1}{\epsilon_1} \right), \epsilon_1 = 8, \quad (15)$$

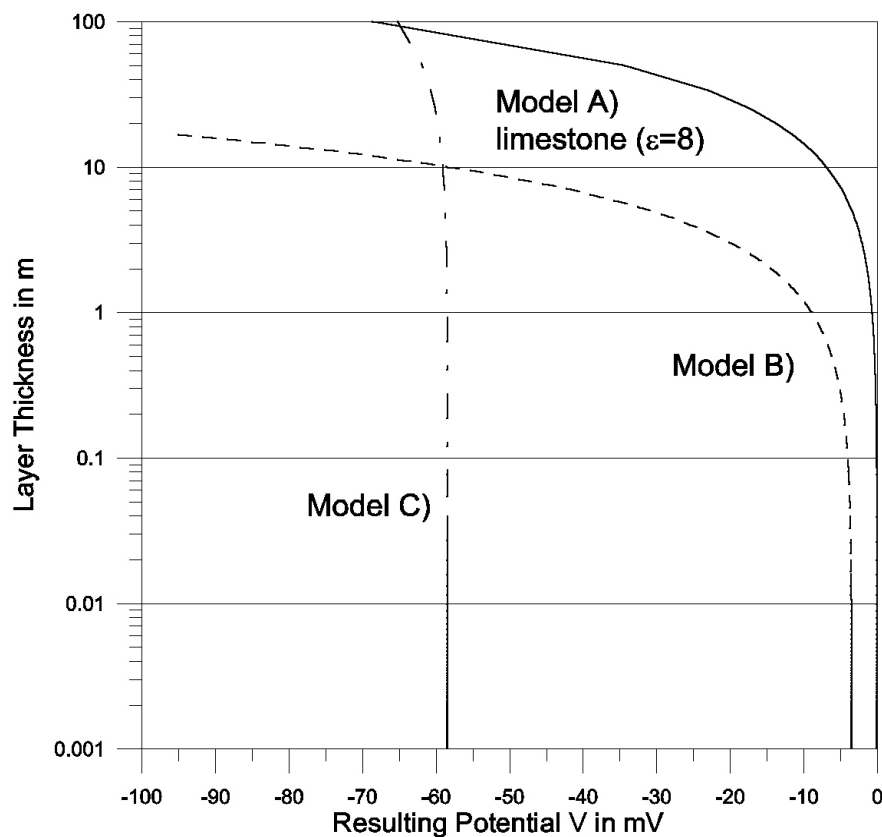
Model B: *limestone and air*

$$V'_{(x,y)} = \frac{D}{\varepsilon_0} \left( \frac{d_1}{\varepsilon_1} + \frac{d_2}{\varepsilon_2} \right), \quad \varepsilon_1 = 8, d_1 = 5m, \varepsilon_2 = 1, \quad (16)$$

Model C: *limestone, air and water*

$$V'_{(x,y)} = \frac{D}{\varepsilon_0} \left( \frac{d_1}{\varepsilon_1} + \frac{d_2}{\varepsilon_2} + \frac{d_3}{\varepsilon_3} \right), \quad \varepsilon_1 = 8, d_1 = 5m, \varepsilon_2 = 1, d_2 = 10m, \varepsilon_3 = 80. \quad (17)$$

For the calculation we used the experimental estimate of the electric displacement  $D = K\varepsilon_0\varepsilon_{\text{limestone}}$ .



**Figure 11:** The resulting potential  $V$  for three specific layered capacitor models.

Model A is a one-layer structure consisting solely of limestone with increasing thickness. Figure 11 shows that this limestone layer will have a measurable effect on the potential distribution at surface with a beginning thickness of some meters.

In Model B, we kept the thickness of the limestone roof constant (5m) and inserted a cavity (air) of varying thickness. This will enhance the measurable effect on the potential distribution at surface.

The strongest effect on the potential distribution is shown in Model C. Here, we studied the wetting effect of a (thin) water layer at the bottom or at the roof of a given cave (roof out of limestone 5m thick, cavity (air) 10m thick). As Figure 11 shows, a thin water layer affects the measurable potential distribution at surface enormously.

From these simple models we can conclude, that an air layer and the wetting effect of water will locally affect the SP values at the surface. Hence, the cave is detectable with the SP technique.

Compared to the model by Aubert et al. (1990) and Aubert & Atangana (1996) our capacitor model is independent from the resistivities of Zone (A) and (B). In analogy to the geo-battery model for large SP anomalies created by geochemical processes (Bigalke & Grabner, 1997) we term our static model a geo-capacitor model.

### *Active model*

Superimposed on the static charge distribution we assume an active charge separation that is controlled by the (micro) climatic conditions in the cave. For shallow cavities, heat and water transport by oscillatory convection plays an important role (Morat et al., 1999). Periodical variations of the temperature, relative humidity, air pressure and the Self-Potential (at the roof of the cave !) are interpreted as manifestations of oscillatory convection motions of the air inside the cave, driven by the geothermal gradient and transporting water as well as heat from the bottom to the roof.

Morat (1999) state that convection is a much more efficient mechanism of heat transport than conduction through the air of the cavity. The air will convect and transport heat, but, at the same time, it will entrain upward the water vapour released at the floor of the cave by evaporation. The vapour will condense on the roof and the upper part of the pillars. In general, the condensed water enters the rock by capillarity, and the relative humidity of the air remains constant in the cave. This aspect together with the streaming potential in the porous media of zone (A) (Revil et al., 1999a, Revil et al., 1999b) plays an important role in our observations.

## **7 Conclusion**

We showed that a shallow cave in karst is detectable with the SP technique. Nevertheless, we expect that the effect of a potential distribution in a deeper structure will be soon attenuated towards the surface and the method will fail. It remains unknown whether the SPS follows the bottom or the roof of the cave. However, from the findings in the field experiment (Fig. 8 and Fig. 9) we can conclude that the SPS technique gives the relief of the bottom of the cave. To answer this question, it is planned to study the potential distribution along a crosscut inside the cave.

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